

Athina Bampzeli

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ABOUT

Athina Bampzeli holds an Integrated Master's degree in Chemical Engineering with first-class honours from Aristotle University of Thessaloniki. She is passionate about research and the dissemination of knowledge, and aspires to pursue a career in academia. Her interests lie in the application of technology to industrial production systems, particularly in computer-aided modeling, simulation, and optimization. She is especially interested in hybrid modeling of bioprocesses, a field she aims to specialize in during her PhD. She strongly values lifelong learning and interdisciplinarity, and she is driven by intellectual curiosity.

EDUCATION

Diploma (Integrated Master) in Chemical Engineering

Oct 2020 - Dec 2025

Department of Chemical Engineering, Aristotle University of Thessaloniki

Thessaloniki, Greece

- 1st Class Honours
- Master Thesis: Modeling, simulation and optimization of integrated milk pasteurization and membrane filtration processes. Supervisor: Prof. Michael Georgiadis. Mark 10/10.

RESEARCH INTERESTS

Mechanistic, data-driven, hybrid modeling

Modeling, simulation and optimization of bioprocesses

Process systems engineering

PUBLICATIONS

Bampzeli, A., Charakleias, E., & Georgiadis, M. C. (2026). Modeling, simulation and optimization of integrated milk pasteurization and membrane filtration processes. *Food and Bioprocess Processing*, 157, 583-599.

<https://doi.org/10.1016/j.fbp.2026.04.006>

RESEARCH EXPERIENCE

Thermal Engineer

Mar 2024 - Jan 2026

AcubeSAT, SpaceDot

Thessaloniki, Greece

- Updated the payload and the system-level geometric/thermal model of the AcubeSAT nanosatellite. Conducted thermal and orbital analyses using Simcenter and ESATAN-TMS software. Hands-on tested the components of the payload to see if they can withstand the thermal stresses during the mission. Trained 2 thermal engineer team members.

Diploma Thesis

Oct 2024 - Dec 2025

Department of Chemical Engineering, Aristotle University of Thessaloniki

Thessaloniki, Greece

- Modeled and simulated the dynamic processes of pasteurization and membrane filtration for the production of concentrated milk using gPROMS FormulatedProducts. Integrated the processes in a holistic model. Evaluated the system via sensitivity and uncertainty analyses and optimized the processes considering safety, quality and cost. Worked both independently and in collaboration with Siemens Digital Industries Software.

Capstone design project

Oct 2024 - May 2025

Department of Chemical Engineering, Aristotle University of Thessaloniki

Thessaloniki, Greece

- Led a team of six on the project “Techno-economic Study for the Production of Propylene from Methanol”. Contributed to process modeling (UniSim), energy integration (GAMS), layout optimization (GAMS), net present value Monte Carlo analysis (MATLAB) and parametric analysis for the distillation columns.

Effervescent tonic beverage

Jan 2024 - June 2024

Department of Chemical Engineering, Aristotle University of Thessaloniki

Thessaloniki, Greece

- Designed and conducted iterative formulation experiments for effervescent coffee tablets, evaluating the effects of ingredient composition on aroma, taste, colour and dissolution to optimize product performance. Collaborated in a four-member team to develop the product for the panhellenic Ecotrophelia competition.

SELECTED PROJECTS

Titer Prediction

Mar 2026

- Developed two predictive data-driven models in Python that estimate the amount of product (Titer) produced in the bioreactor during cell-culture experiments. Performed exploratory data analysis, data preprocessing, model training, testing, evaluation and visualization of the results.

Programming Languages Course

May 2025

- Developed and evaluated machine learning and surrogate models in Python for predicting bioconcentration factor (BCF). Applied dimensionality reduction, clustering, hyperparameter tuning and surrogate-based optimization.

Advanced Methods for the Design, Synthesis and Optimization of Processes

Oct 2023 - Dec 2023

- Developed MILP optimization models in GAMS within a two-member team for four projects, including steady-state supply chain network optimization, state-task network scheduling, heat exchanger network synthesis and energy planning.

SELECTED SKILLS

Technical skills: gPROMS, Unisim, Python, GAMS, MATLAB, Fortran, ESATAN-TMS, Ubuntu, Virtual Machines

Soft skills: teamwork, effective communication, independence, organization, curiosity, interdisciplinary collaboration, creativity, critical thinking, problem solving, adaptability, resilience, discipline, leadership, project management, attention to detail, ethical research

CONFERENCES & SEMINARS

AcubeSAT Payload EQM TVAC Test Campaign, Participant

Oct 2025

ESA FYS! 3

ESEC-Galaxia, Transinne, Belgium

Space Standards Training Course 2025, Participant

June 2025

ESA Academy

ESEC-Galaxia, Transinne, Belgium

Cybersecurity and Mathematics, Speaker

May 2025

15th International Week Dedicated to Maths 2025

Thessaloniki, Greece

Cybersecurity Seminar, Participant

Dec 2024 - May 2025

Aristotle University of Thessaloniki - Google.org

Thessaloniki, Greece

Uncertainty Quantification in Numerical Simulations, Participant

Nov 2024

École des Ponts ParisTech

Paris, France

Optimization Methods for Energy Systems Engineering, Participant

July 2024

Princeton Athens Center

Athens, Greece

14th Panhellenic Scientific Conference on Chemical Engineering, Attendee

May 2024

Aristotle University of Thessaloniki

Thessaloniki, Greece

ADDITIONAL PROFESSIONAL EXPERIENCE

Cybersecurity Intern

Oct 2025 - Dec 2025, Full time

Olympia Electronics

Thessaloniki, Greece

- Developed presentations on cybersecurity topics, focusing on cyber threats, security tools, and best practices for individuals and enterprises. Completed incident response simulation exercises covering phishing/data breaches, ransomware, and insider threats, focusing on attack detection, containment, system recovery, incident communication, and long-term prevention strategies. Designed phishing awareness campaigns using Gophish, gaining hands-on experience with phishing techniques, email indicators of compromise, and employee response evaluation within an organizational context. Conducted penetration testing in a virtualized environment using Kali Linux (attacker) and Metasploitable (victim), including vulnerability assessment with Nessus.

Chemistry Tutor

Sep 2023 - May 2024, Part time

Tutoring Center

Thessaloniki, Greece

- Taught chemistry to first- and second-year high school students, organized the teaching materials, and designed their assessment and examination methods.

VOLUNTEERING

Board of European Students of Technology (BEST) Thessaloniki

Nov 2021 - Mar 2024

- Member of the Human Resources Team (July 2023 - Mar 2024)
- Vice President for Corporate and Public relations (July 2022 - June 2023)
- Logistics Coordinator for cultural exchange event (Apr 2022 - June 2022)
- Member of the Public Relations, Corporate Relations and Design Team (Nov 2021 - June 2022)

AWARDS & ACHIEVEMENTS

Admitted first to the Department of Chemical Engineering

June 2020

Aristotle University of Thessaloniki

Greece

One of the 24 Greek student representatives in the Euroscola program

Feb 2019

European Parliament

Strasbourg, France

Qualification for the 34th National Mathematical Olympiad “Archimedes”

Mar 2017

Hellenic Mathematical Society

Athens, Greece

12th Prize in the 4th Panhellenic Physics Competition

June 2016

Association of Greek Physicists

Thessaloniki, Greece

LANGUAGES

Greek: Native

English: C2 - University of Michigan (2017)

German: B2 - ÖSD certificate (2018)